Drew University - CSCI 260 - Fall 2024
HW4 - LC-3 Assembly Language

HW Info.: Due on 10/25/2024 (F) by 11:59pm, 37 points
Submission: Submit as a single PDF file on Moodle.
Note: Please follow Drew University's Standard of Academic Integrity and refer to the course syllabus for late submission policy.

1. (15 points) Write an LC-3 program that utilizes R1 to count the number of 1s appeared in R0. For example, if we manually set R0 = 0001001101110000, then after the program executes, we should have R1 = #6.

   In order to get full credits, you need to give me (1) a flow chart on the program you design, and then (2) the assembly code of it, and lastly (3) the code converted to hexadecimal.

2. (12 points) Given that every register is initialized to zero, show the contents of the registers in hexadecimal after the following segment of code is fully executed.

Address	*Label*	*Instruction*	
0xA400	THIS1	LEA	R0, THIS1
0xA401	THIS2	LD	R1, THIS2
0xA402	THIS3	LDI	R2, THIS5
0xA403	THIS4	LDR	R3, R0, #2
0xA404	THIS5	.FILL	xA400

3. (10 points) For the program below, describe what does it do using at most 20 words, and find the value stored at the memory location labelled with RESULT after the program terminates.

Label	*Instruction*	
	.ORIG	x3000
	LD	R2, ZERO
	LD	R0, M0
	LD	R1, M1
LOOP	BRz	DONE
	ADD	R2, R2, R0
	ADD	R1, R1, #-1
	BRnzp	LOOP
DONE	ST	R2, RESULT
	HALT	
RESULT	.FILL	x0000
ZERO	.FILL	x0000
M0	.FILL	x0004
M1	.FILL	x0803
	.END	